

ETHIOPIAN PAY-LATER PLATFORM

Software Requirements Specification (SRS)

IEEE-style technical requirements baseline

Ethiopia-designed • Addis Ababa-first

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CONFIDENTIAL WORKING DOCUMENT

Document Control

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1.0	21 August 2026	Eleanor Tefera (FinTech)	Working baseline	Initial researched package release

Research Status Legend

VERIFIED	Directly supported by an official/primary source.
REPORTED	Supported by a reputable third-party source.
INFERENCE	Reasonable conclusion from available evidence.
ASSUMPTION	Design assumption for planning; not presented as market fact.
REQUIRES CONFIRMATION	Needs provider, bank, NBE, legal or business-partner confirmation.

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1. Introduction

This SRS specifies functional and non-functional requirements for the Ethiopian Pay-Later platform. It is implementation-neutral where practical and deliberately separates internal requirements from unconfirmed external-provider capabilities.

2. Purpose

Provide a testable contract between product, engineering, QA, security, finance operations and compliance before coding.

3. Scope

- Customer mobile application
- Merchant mobile application
- Admin web platform
- Enterprise merchant integration
- Credit/eligibility and installment services
- Payment-provider abstraction and reconciliation
- Merchant settlement/refunds
- Risk, fraud, disputes, support and audit

4. Definitions

Term	Definition
BNPL	Buy Now, Pay Later purchase financing/credit product.
First Payment	Required customer payment at/near purchase before financing activation/release under applicable plan.
Available Limit	Approved exposure minus reserved/used/restricted amounts according to policy.
PENDING/UNKNOWN	External payment outcome not yet authoritative; not equivalent to failure.
Settlement	Process of paying merchant according to commercial/funding model.
Reconciliation	Comparison of internal financial records with provider/bank/settlement evidence.
Idempotency	Guarantee that safe retries do not create duplicate business/financial effects.
KYB	Know Your Business / merchant verification.
Ledger	Immutable balanced financial record of money/obligations.

5. System Overview

Customer App + Merchant App + Admin + Enterprise Merchants → Backend → Identity/KYC • Credit/Risk • Orders/BNPL • Payments • Ledger/Settlement • Support → External credit, payment, identity and notification partners

6. Users and Roles

Role	Allowed Scope	Restriction
Customer	Use Pay-Later, payments, history, support	No merchant/admin access
Merchant Owner	Manage merchant, branches, staff, sales, refunds/reports	Cannot approve platform credit or edit settlements
Merchant Manager	Operational merchant management	No ownership/legal-profile changes unless delegated
Merchant Employee	Create/view branch sales and QR	No settlement/configuration access by default

Finance Officer	Payments, reconciliation, settlements, refunds	No KYC approval or risk-policy publishing
Risk Officer	Eligibility/risk review, limits, fraud review	No settlement execution
Compliance Officer	KYC/KYB, complaints, regulatory evidence	No unrestricted financial posting
Customer Support	Tickets, assisted troubleshooting	Masked sensitive data; no credit/settlement overrides
Operations Officer	Provider incidents, transaction operations	No unrestricted policy/role changes
Super Administrator	Emergency/approved platform administration	Break-glass controls, MFA, enhanced audit
System Administrator	Technical settings and platform health	Cannot approve financial business decisions

7. Functional Requirements

Requirement ID	Requirement
BR-001	The platform shall register customers using a verified Ethiopian mobile number and OTP.
BR-002	The platform shall complete required KYC before any Pay-Later credit is activated.
BR-003	The platform shall treat identity verification and credit eligibility as separate decisions.
BR-004	A first-time customer shall receive a documented first-time eligibility decision before any financed exposure.
BR-005	First-time outcomes shall support starter limit, transaction-only approval, higher down-payment offer, more information, manual review, decline and temporary block.
BR-006	Every purchase shall be subject to transaction-level risk/eligibility checks even when a customer has an available limit.
BR-007	Where a first payment is required, merchant release shall not occur until that payment is authoritatively SUCCESS.
BR-008	The platform shall support both large enterprises and small/local merchants.
BR-009	Small merchants shall be able to create Quick Sales without maintaining a catalog or POS.
BR-010	The platform shall support physical and online commerce.
BR-011	The platform shall support QR-assisted physical checkout, aligned where feasible with Ethiopian interoperability standards.
BR-012	The platform shall support multiple Ethiopian payment channels only where legal, commercial and technical capability is confirmed.
BR-013	No provider-specific API capability shall be represented as available without evidence or partner confirmation.
BR-014	Customers shall see all installment obligations in one platform regardless of supported repayment rail.
BR-015	Installment plans, down payments, frequencies and eligibility rules shall be configurable and versioned.
BR-016	Before acceptance, customers shall see purchase price, first payment, financed amount, installment schedule, fees and total payable.
BR-017	The platform shall support early/full repayment and optional partial prepayment according to product policy.
BR-018	The platform shall issue payment reminders and support safe retries.
BR-019	PENDING or UNKNOWN payment status shall never be treated as FAILED without authoritative evidence.
BR-020	The platform shall support full and partial refunds with installment and ledger adjustment.
BR-021	Merchant settlement shall be traceable from gross sale to net settlement and adjustments.
BR-022	Every money movement shall be represented by immutable financial records/double-entry ledger postings.
BR-023	Customers and merchants shall have defined dispute and complaint workflows.
BR-024	Complaint records shall preserve SLA evidence including NBE escalation timing where applicable.
BR-025	Merchants shall complete business verification and risk assessment before activation.

BR-026	Fraud controls shall cover duplicate identities, device abuse, account takeover, fake merchants, collusion and suspicious refunds.
BR-027	Overdue behavior shall trigger policy-driven reminders, restrictions and review; aggressive collection shall not be automatic.
BR-028	The system shall support hardship/manual arrangements and future restructuring subject to lender/legal policy.
BR-029	Privileged access shall follow least privilege, role separation and strong administrator controls.
BR-030	Important credit, financial, KYC, configuration and admin actions shall be auditable.
BR-031	Personal data shall be collected for defined purposes, protected and retained/deleted under Ethiopian requirements.
BR-032	The payment architecture shall remain provider-independent and support gateway, direct or hybrid integration strategies.
BR-033	Critical customer-facing content shall support English and Amharic for the Addis Ababa launch.
BR-034	Credit approval and payment authorization shall require online confirmation unless a separately approved risk model explicitly permits otherwise.
BR-035	Management shall receive reporting on GMV, active users/merchants, repayment, overdue, fraud, refunds, settlements and revenue.
BR-036	Fees shall be configurable; customer/late/financing charges shall not be enabled without legal/product approval.
BR-037	The legal creditor/funder/receivable owner shall be configurable by product/partner model rather than hard-coded as the platform company.
BR-038	Payment-provider strategy shall permit gateway-first MVP and selective direct integrations later.
BR-039	Terms, disclosures and consent acceptance shall be versioned and retained as evidence.
BR-040	The first production release shall use a controlled Addis Ababa pilot with limited merchants, limits and plans.

8. Non-Functional Requirements

ID	Requirement
NFR-001	Financial calculations shall use integer minor units or exact decimal types; binary floating point is prohibited for authoritative monetary values.
NFR-002	Financial write operations shall be transactional and idempotent.
NFR-003	The system shall preserve an immutable audit trail for credit, payment, refund, settlement, configuration and privileged actions.
NFR-004	Core APIs shall return correlation IDs and structured error codes.
NFR-005	Provider outages shall degrade gracefully without corrupting local state.
NFR-006	Customer/merchant mobile interfaces shall be usable on low-end Android devices and slow networks.
NFR-007	Sensitive data shall be encrypted in transit and at rest, with secrets stored outside source code.
NFR-008	The architecture shall support horizontal application scaling and independent worker scaling.
NFR-009	Critical screens shall distinguish loading, pending, success, failure and empty states.
NFR-010	English and Amharic content shall be supported for critical customer/merchant journeys in the pilot.

9. External Systems

System	Verified Capability	Integration Status
Telebirr	Public C2B/H5 and schedule-payment documentation.	Direct integration feasible only after merchant/commercial onboarding; BNPL mandate use requires confirmation.
CBE Birr	Goods purchase/bill payment publicly documented.	Direct API/sandbox not publicly confirmed in reviewed sources; use gateway/partnership if confirmed.

M-PESA Ethiopia	Merchant payments/business onboarding publicly documented.	Ethiopia-specific BNPL API/sandbox requires confirmation; do not use Kenya assumptions.
Payment gateways	NBE lists multiple licensed/commercialized operators.	Evaluate coverage, onboarding, callbacks, refunds and settlement commercially.
Fayda / identity	Relevant national identity direction.	Exact integration, consent and data-use model requires confirmation.
Credit/funding institution	Required by selected legal/business model.	Interface must be partner-agnostic.

10. Data Requirements

- ULID/UUID-style non-sequential public identifiers; database internal choice documented.
- Foreign-key integrity for critical relations.
- Unique provider reference/idempotency keys where applicable.
- State constraints and version numbers for critical records.
- PII/KYC data minimized and access-controlled.
- Financial histories append/reverse; never silently overwrite posted records.

11. Security Requirements

- Strong authentication, OTP abuse limits and device/session controls.
- RBAC and separation of duties for finance/risk/compliance/admin.
- MFA for privileged roles.
- Webhook signature/allowlist validation where provider supports it; replay protection always.
- Rate limiting, brute-force protection and suspicious-session detection.
- No raw card credentials stored unless a separately approved PCI-compliant model requires it.

12. Financial Requirements

- Double-entry ledger with balanced transaction entries.
- Credit reservation to prevent concurrent overspend.
- First payment and financed exposure represented separately.
- Refunds and reversals use linked adjustment records.
- Merchant payable remains open on settlement failure.
- Every provider/bank posting reconciled to unique evidence.

13. Payment Requirements

State	Meaning / Rule
CREATED	Internal intent created; no provider action completed.
PROCESSING	Provider interaction in progress.
PENDING / UNKNOWN	Outcome unresolved; no failure assumptions.
SUCCESS	Authoritatively successful; financial posting allowed once.
FAILED	Authoritatively failed; no success posting.
CANCELLED	Cancelled before completion where permitted.
PARTIALLY_REFUNDED	Some successful value reversed/refunded.
REFUNDED	Fully refunded according to approved process.
DISPUTED	Dispute flag/workflow; original financial history remains.

14. Compliance Requirements

- Version terms/consents/disclosures.
- Support complaints and evidence retention.

- Implement data subject rights and retention based on Ethiopian legal interpretation.
- Maintain configurable legal creditor/funder ownership.
- Do not hard-code Qatar limits, fees, age/residency or retention rules.

15. Reporting Requirements

- Customer statement and upcoming installments.
- Merchant sales, fees, refunds and settlements.
- Management GMV, exposure, approval, repayment, overdue, fraud, reconciliation and revenue.
- Compliance KYC/KYB, complaints, suspicious activity and audit extracts.

16. Error Handling

Class	Expected Behavior
Validation	4xx structured error with field details; no state change.
Conflict / duplicate	Return prior result or conflict reference; no duplicate financial effect.
Provider timeout	Set PENDING/UNKNOWN and reconcile; do not assume failure.
Unauthorized	Reject, audit sensitive attempts and avoid data leakage.
Internal failure	Rollback transaction, emit correlation ID, alert if critical.
Settlement failure	Keep payable outstanding, record failure and retry under finance controls.

17. Audit Requirements

Audit entries capture who, what, when, previous/new state, reason, source/device, correlation ID and policy/config version for significant actions. Audit evidence is immutable to ordinary users.

18. Availability Requirements

- Pilot target: design for high availability of customer/merchant checkout and repayment services; exact SLA finalized with infrastructure/provider contracts.
- Read-only history may remain available during some external-provider outages.
- Provider health status must be visible to operations.

19. Performance Requirements

Operation	Initial Target (ASSUMPTION)
Login / profile API	p95 < 1.5 s excluding OTP/provider latency
Eligibility decision	p95 < 3 s when no external/manual dependency; otherwise explicit pending/review
Checkout quote	p95 < 2 s excluding external decision latency
Payment initiation	p95 < 2 s before provider handoff
Admin common views	p95 < 3 s for bounded queries
Reports	Asynchronous for large datasets; progress/export notification

20. Scalability Requirements

- Stateless application nodes behind load balancer.
- Redis for cache/locks/rate limiting and queues as justified.
- Dedicated workers for payments, notifications, reconciliation and reports.
- PostgreSQL indexing/partitioning/archival as transaction volume grows.
- Provider adapters independently rate-limited and circuit-broken.

21. Backup and Recovery

- Encrypted automated database backups and point-in-time recovery where supported.
- Object/KYC file backup/versioning according to retention policy.
- Initial target RPO ≤ 15 minutes and RTO ≤ 4 hours for core financial database, subject to production architecture and business approval.
- Quarterly restore exercises and documented disaster-recovery runbook.

22. Acceptance Criteria

- All BR/NFR requirements mapped to test evidence.
- No Sev-1/Sev-2 unresolved security/financial defects before pilot.
- Payment duplicate/delay/pending scenarios proven in test.
- Ledger balances and reconciliation test fixtures pass.
- Provider capabilities used in production have written/official confirmation.
- Compliance/legal Phase-0 decisions recorded and reflected in configuration/agreements.

References

- **[ETH-NBE-01] VERIFIED — National Bank of Ethiopia.** Payment Instrument Issuers / System Operators.
<https://nbe.gov.et/payment-instrument-issuers-system-operators/>
Supports: Current licensed/commercialized payment instrument issuers and payment system operators, including Telebirr, Kacha, M-PESA Ethiopia, EthSwitch, ArifPay, Chapa and SantimPay.
- **[ETH-NBE-02] VERIFIED — National Bank of Ethiopia.** Financial Stability Report — March 2026.
https://nbe.gov.et/wp-content/uploads/2026/03/Financial-Stability-Report_MARCH_2026.pdf
Supports: Current payment-sector regulatory framework, digital-finance context and systemic risk context.
- **[ETH-NBE-03] VERIFIED — National Bank of Ethiopia.** National Digital Payments Strategy 2026–2030 — Draft.
https://nbe.gov.et/wp-content/uploads/2025/12/Ethiopia_NDPS_Draft_F.pdf
Supports: Interoperability direction and digital-payment strategy context; document is a draft.
- **[ETH-NBE-04] VERIFIED — National Bank of Ethiopia.** Financial Consumer Protection — Complaint Handling.
<https://nbe.gov.et/fcpe/complaint/>
Supports: Complaint escalation after first contacting the financial service provider; NBE review where unresolved/no response within 10 business days.
- **[ETH-DP-01] VERIFIED — Federal Democratic Republic of Ethiopia / Ministry of Justice.** Personal Data Protection Proclamation No. 1321/2024.
<https://justice.gov.et/en/law/personal-data-protection-proclamation/>
Supports: Ethiopian personal-data protection framework.
- **[ETH-QR-01] VERIFIED — EthSwitch.** Interoperable QR Standard.
<https://ethswitch.com/wp-content/uploads/2024/12/Interoperable-QR-Standard.pdf>
Supports: Interoperable QR design direction for merchant payments and limitations of closed-loop QR.
- **[TEL-01] VERIFIED — Ethio telecom.** Telebirr Developer Portal — Getting Started.
<https://developer.ethiotelecom.et/docs/GettingStarted>
Supports: Publicly documented Telebirr business integration categories including C2B, B2B, H5, Mini App and subscription payment.
- **[TEL-02] VERIFIED — Ethio telecom.** Telebirr H5 C2B Web Payment Integration.
<https://developer.ethiotelecom.et/docs/category/h5-c2b-web-payment-integration>
Supports: Public create-order, checkout, query/status, callback and refund integration documentation.
- **[TEL-03] VERIFIED — Ethio telecom.** Telebirr Schedule Payment / CreateDisburseOrder.
<https://developer.ethiotelecom.et/docs/Schedule%20Payment/CreateDisburseOrder>
Supports: Public documentation for mandate-based scheduled deduction; BNPL repayment use requires commercial/regulatory confirmation.
- **[CBE-01] VERIFIED — Commercial Bank of Ethiopia.** CBE Birr.
<https://combanketh.et/ways-of-banking/cbe-birr>
Supports: Official description of CBE Birr customer/mobile-money services.
- **[CBE-02] VERIFIED — Commercial Bank of Ethiopia.** CBE Products and Services.
https://combanketh.et/cbeapi/uploads/CBE_Products_and_services_English_84e511c740.pdf
Supports: CBE Birr supports transfers, cash out, airtime, goods purchase and bill payment; no equivalent public direct BNPL integration specification verified.
- **[MPE-01] VERIFIED — Safaricom Ethiopia.** M-PESA Ethiopia Merchant Terms / Business Use.
<https://www.safaricom.et/>
Supports: M-PESA Ethiopia supports merchant payments/business arrangements; exact BNPL API onboarding and sandbox require provider confirmation.
- **[QTR-TC] VERIFIED — PayLater Qatar.** PayLater General Terms and Conditions — client-provided 20-page copy.
Client-provided PDF reviewed 21 August 2026

Supports: Customer/merchant/PayLater roles, eligibility, 25% first payment, PayMore, limits, repayment, default, restructuring, disputes, data handling and account rules.

- **[QTR-PL-01] VERIFIED — PayLater Qatar.** PayLater Terms and Conditions.

<https://paylaterapp.com/terms-and-conditions/>

Supports: Current public PayLater Qatar contractual product model.

- **[QTR-PL-02] VERIFIED — PayLater Qatar.** How can I sign up for PayLater?.

<https://help.paylaterapp.com/en/support/solutions/articles/154000139292-how-can-i-sign-up-for-paylater->

Supports: Phone OTP, QID capture and liveness-based customer onboarding.

- **[QTR-PL-03] VERIFIED — PayLater Qatar.** What is my credit limit and how much can I spend?.

<https://help.paylaterapp.com/en/support/solutions/articles/154000139303-what-is-my-credit-limit-and-how-much-can-i-spend-with-paylater->

Supports: Credit limit uses creditworthiness, PayLater repayment history and account usage.

- **[QTR-PL-04] VERIFIED — PayLater Qatar.** Where can I use PayLater?.

<https://help.paylaterapp.com/en/support/solutions/articles/154000139291-where-can-i-use-paylater->

Supports: Use across online and in-store merchants.

- **[QTR-CBQ-01] VERIFIED — Commercial Bank Qatar / Vodafone Qatar.** Vodafone Easy-Buy / CBQ Buy Now Pay Later.

<https://www.vodafone.qa/en/buynowpaylater>

Supports: Bank-card installment model for online and selected physical stores; full purchase reduces card limit and customer repays equal monthly installments.

- **[QTR-VF-01] VERIFIED — Vodafone Qatar.** Vodafone Partners with Commercial Bank to Offer BNPL.

<https://www.vodafone.qa/en/about-us/media/press-release/vodafone-partners-with-commercial-bank-to-offer-buy-now-pay-later-installment-plan>

Supports: Qatar device-installment model through bank partnership and merchant stores.